Hypergeometric

Recall: The system works if at least 1 of the 6 components works. ($p = 0.68$)

What is $P(X \geq 1)$? (The probability the system works)

$P(X \geq 1) = P(X = 1) + P(X = 2) + \ldots + P(X = 6)$

Possible values of X:

$$\underset{X=0}{0}, \underbrace{1, 2, 3, 4, 5, 6}_{X \geq 1}$$

$$P(X \geq 1) = 1 - P(X = 0)$$

(We are choosing 7 people from 32 men and 26 women)

What is the probability the committee contains at least 6 women?

$$P(X \geq 6) = P(X = 6) + P(X = 7)$$

$$= \frac{\binom{26}{6}\binom{32}{1}}{\binom{58}{7}} + \frac{\binom{26}{7}\binom{32}{0}}{\binom{58}{7}}$$

$$= 0.0245 + 0.0022$$

$$= 0.0267$$

Credit to the owner